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Automating MARC Holdings Generation with AI: Lessons from the Field

Built with an AI coding assistant; runs without one.

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More detail, the figures and a PDF of this poster
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1 The problem, and the approaches I tried

The problem

Many libraries' serials holdings are stored only as human-readable, free-text **866** fields rather than as MARC-encoded **853/863** pairs. Converting by hand is slow, conventions vary, and hand-typed data is prone to typos and other errors.

> **The goal:** a tool that is portable across collections, works on messy holdings as they are (no manual clean-up), and never silently loses anything.

866 · as a person typed it

```
v. 27 no. 4-v. 31 no. 4
a  a  b  b  a  a  b  b

(April 1992-April 1996)
 j     i     j     i
```

↓

subfield	853 caption	863 held
$a volume	v.	27-31
$b issue	no.	4-4
$i year	(year)	1992-1996
$j month	(month)	04-04

```
853 31 $8 1 $a v. $b no. $i (year) $j (month)
863 40 $8 1.1 $a 27-31 $b 4-4 $i 1992-1996 $j 04-04
```

APPROACH 1

Write the script by hand

I adapted USC colleagues' script (Chung & Pendyala, 2023) over six months, teaching myself Python. The hardest part was one monolithic regular expression: weeks, if not months, in regex101 before it covered every pattern.

Pros: tuned to one collection; code I knew line by line.

Cons: only feasible on clean, consistent 866s, and not portable. Few institutions could write their own expression that way.

APPROACH 2

Ask a model to read the statements

Send the 866s to a large language model through its API, and ask it to pick out the volumes, issues and dates an 853/863 is built from.

Pros: the least coding knowledge; good results from minimal prompting.

Cons: tokens for every statement (free access stopped within two dozen records), and output that isn't guaranteed to be the same twice.

APPROACH 3 · CHOSEN

Use AI to build the tool

Describe in plain language what the tool should do, and have an AI coding assistant (Claude Code) write a deterministic Python application.

Pros: the same file converts the same way every time, with no model, API key or tokens when it runs.

Cons: a black box. I can confirm it works, and tests stop it regressing, without fully understanding why it works; and it may be overengineered, which I can't judge.

	1 · BY HAND	2 · A MODEL READS	3 · AI BUILDS THE TOOL
Coding knowledge	The most	The least	Some
Cost	Nothing	Tokens for every statement	A pro-level AI assistant to build; nothing to run
Same output every time	Yes	Not guaranteed	Yes
Other collections	Poorly	Not tested far enough	Designed for it
Understanding the code	Fully	No code to read	Partly: a black box, kept honest by tests

2 The approach chosen: AI-assisted development of a deterministic tool

How the Workbench reads a statement

1 · Cut into labelled pieces

```
v.  27  no.  4   -   v.  31  no.  4
VOL  #  ISS  #   –  VOL  #  ISS  #

(  April  1992  -  April  1996  )
     CHRON  YEAR  -  CHRON  YEAR
```

2 · Its shape

```
VOL ISS – VOL ISS(CHRON YEAR-CHRON YEAR)
```

```
v. <start_vol> no. <start_iss>-v. <end_vol> no. <end_iss>
(<start_month> <start_year>-<end_month> <end_year>)
```

1. Statements with the same shape are one group, whatever their numbers.
2. Each group gets a matching rule with a named slot for each value.
3. A cataloguer confirms what each value means, once per group.
4. The parser reads every statement; a pattern fills only what it cannot know.

What the cataloguer controls

- **On a pattern:** what each value is, its caption and level; skip a shape; test or edit the rule; reuse the library on another collection.
- **On one record:** edit an 866 before it is read, split it, set this record's 853 (frequency, numbering, issues per volume, missing captions), skip it, leave a note.
- **For the whole file:** holdings level, subfield convention, strict reading, what to do with a value MARC has no place for, and whether to keep each 866.

Every decision is listed in a downloadable log.

The working rules

1. **Measure before you change.**
2. **Nothing disappears silently.**
3. **One reader of holdings.**
4. **If a fix needs the tests rewritten, stop.**
5. **Write the changelog for cataloguers.**

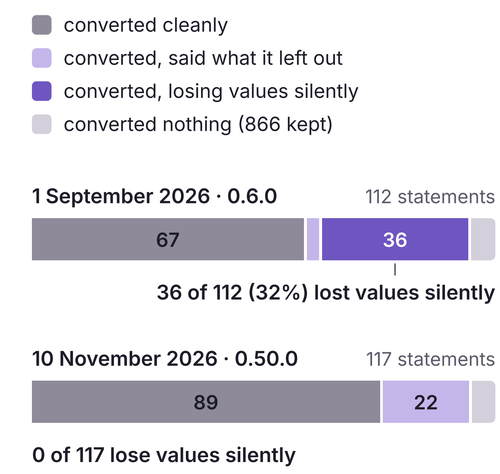
The human in the loop

The AI writes the code; the cataloguer decides what "correct" means. Two bugs were found by the cataloguer, not the tests:

- `v. 58 Suppl. (Sep 2003)` lost "Suppl." when its pattern was confirmed (fixed in 0.32.0).
- `Late Summer 2002` was coded as plain Summer, though there may be an Early Summer issue too; now it is flagged (fixed in 0.9.6).

The turning point: silent losses

A silent loss writes the fields, removes the 866, and drops a volume or a month with nothing on screen to say so.

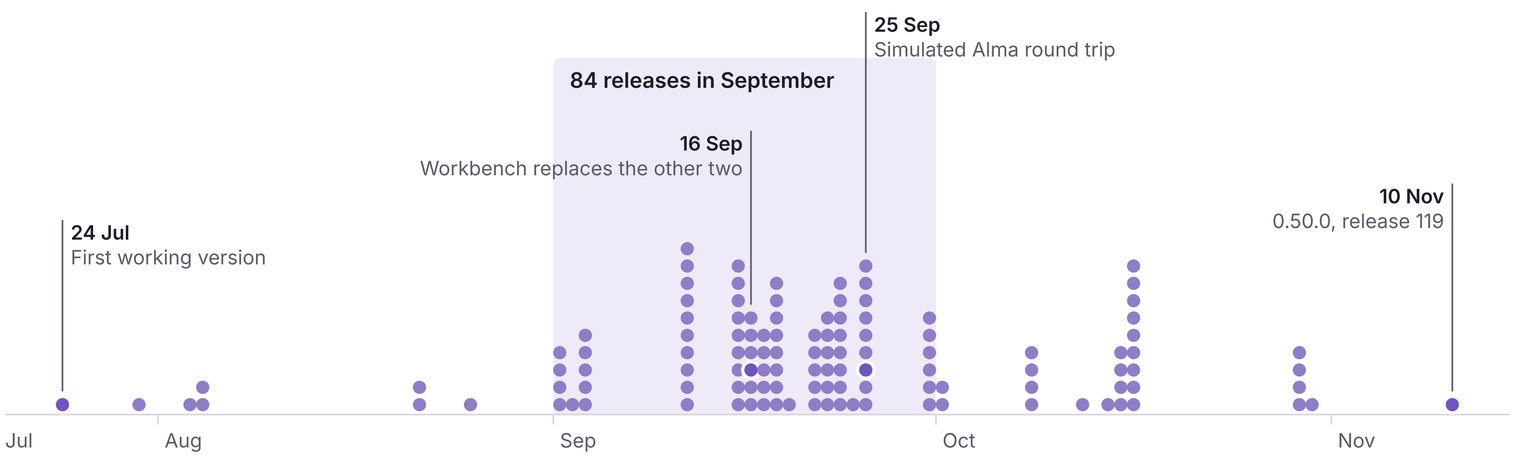


On real data

A real Alma export of 372 records:

1,057 holdings statements	**0** values unaccounted for
0 drift, simulated Alma round trip	**48/48** of Alma's own 866s reproduced

How it was built: 119 numbered versions in 15 weeks, most of them small



Each dot is one version. The Pattern Detector and Converter began as one-shot prompts; the Workbench combined them, and on 16 September replaced them.

Lessons, and what's next

- **A script reads what it was written for, and passes over the rest.** The Workbench makes the check part of the tool: every value is encoded, named as left out, or held with its 866 kept.
- **Decide whether you can work with code you don't fully understand.** A test suite and automated checks are what made it acceptable.
- **AI is at its best building the tool, not being it.**
- **Next:** testing on more real .mrc files from technical services colleagues, and a review by a software engineer, working toward a 1.0 release.

Try it: tools.matthewcodey.com/workbench · **Source:** github.com/bdmcodey/holdings-workbench

Chung, M., & Pendyala, P. C. (2023). Enhancing serials holdings data: A Pymarc-powered clean-up project. *Code4Lib Journal*, 58.